Flock Assignment

Smart-Meter Telemetry Cleaning & NILM Appliance Disaggregation (UK REFIT Dataset)
GitHub Repository: https://github.com/devank-k/Flock_energy_assignment

1. Executive Summary

This submission covers the complete take-home assignment on the UK REFIT electrical smart-meter dataset. The project is structured into three main sections: raw telemetry auditing and 1-minute resampling (House 1), washing machine cycle detection heuristics across four homes (Houses 1, 2, 3, and 5), and causal Non-Intrusive Load Monitoring (NILM) model training and evaluation using LightGBM. All code, cleaned datasets, and generated visualizations are reproducible end-to-end via python run_pipeline.py.

2. Raw Telemetry Audit & Resampling Pipeline

Auditing the raw 8-second sensor readings for House 1 (4,377,877 records between October 2013 and May 2015) revealed several significant real-world telemetry irregularities:

- **Timestamp duplicates:** Found 645,514 duplicate timestamp strings and 1,004,404 duplicate Unix epochs, caused by transceiver burst retries. These were resolved by binning into uniform intervals and averaging values.
- **ADC register overflows:** 3,091 sensor spikes exceeded physically possible household limits. Specific spike values matched 16-bit register limits exactly: 32,767 (2^{15}-1), 65,535 (2^{16}-1), and 98,301. Aggregate mains power was capped at 20 kW and sub-metered appliance lines were capped at 4 kW.
- **Dropouts vs. Outages:** Sampling gaps were split into short dropouts (≤ 5 minutes) and extended power outages (> 5 minutes). Dropouts accounted for 0.12% of the timeline and were safely interpolated. Extended outages accounted for 14.86% of the timeline (dominated by a 41.6-day system outage in February 2014) and were flagged with is_outage=True and set to zero rather than interpolated, preventing synthetic power injection into the data.
- **Final Grid:** The resampled 1-minute dataset contains 513,322 timestamps, with 85.02% representing valid recorded data.

3. Washing Machine Cycle Detection & Cross-Household EDA

A washing machine has a complex multi-phase power signature: an initial 2.0 to 2.4 kW resistive heating spike (warming water for 8-15 minutes), alternating 250-450 W motor tumble and rinse phases, and periodic zero-power soak pauses. A simple power threshold fails because it treats soak periods as the end of a cycle, artificially fragmenting one wash into three or four separate events.

We implemented a deterministic physics-based state machine: standby threshold of 5 W, active threshold of 150 W (to catch mechanical tumbling), a 10-minute maximum gap allowance (to bridge soak pauses within the same wash), and a minimum requirement of 15 minutes duration and 0.10 kWh energy to filter out spurious door opens or electronic wake-ups.

Applying this heuristic across four households with washing machines yielded clear behavioral benchmarks:

- **House 1 (Beko, 2 occupants):** 377 cycles detected. Median cycle duration was 33.0 minutes, consuming 0.321 kWh per cycle, with a peak draw of 2,327 W. Standby/inactive time was 98.36%.
- **House 2 (LG Direct Drive, 4 occupants):** 323 cycles detected. Median duration was 97.0 minutes, consuming 0.621 kWh per cycle, with a peak draw of 2,090 W. Reflects preference for longer eco-wash cycles. Standby time was 96.47%.

- **House 3 (UK Front-loader, 2 occupants):** 641 cycles detected. Median duration was 66.0 minutes, consuming 0.558 kWh per cycle, with a peak draw of 1,927 W. Standby time was 94.98%.
- **House 5 (AEG Lavamat, 4 occupants):** 718 cycles detected. Median duration was 48.5 minutes, consuming 0.309 kWh per cycle, with a peak draw of 2,072 W. Standby time was 92.84%.

Across all four homes, washing machines spend 93% to 98% of their time completely idle. Start-time distributions show clear diurnal patterns: concentrated usage on weekend mornings (08:00 to 11:00) and secondary peaks in mid-afternoon.

4. NILM Disaggregation Modeling & Evaluation

Validation Setup: Evaluated on House 1 using a strict chronological split (70% train: Oct 2013 - Nov 2014, 15% val: Nov 2014 - Jan 2015, 15% test: Jan 2015 - May 2015; 120,566 test minutes). Grid outage periods were masked out. To ensure zero lookahead data leakage for online meter deployments, all feature engineering used strictly backward-looking causal rolling windows (15m, 60m, 240m means, standard deviations, min/max, and lagged deltas) plus cyclic time features.

We evaluated four models on the out-of-sample test set:

- **Always-Off Baseline:** MAE 11.04 W, RMSE 137.03 W, Normalized Disaggregation Error (NDE) 1.000, Total Energy Error 100%, F1 0.000. This illustrates the *sparsity paradox*: MAE appears low simply because the appliance is off 98% of the time, even though it detects none of the actual energy.
- **Hourly Mean Profile Baseline:** MAE 19.80 W, RMSE 136.04 W, NDE 0.986, Total Energy Error 12.28%, F1 0.127. Predicts the historical average draw for each hour of the week.
- **Direct LightGBM Regressor:** MAE 11.14 W, RMSE 98.21 W, NDE 0.514 (a 48.6% reduction in error over baselines), Total Energy Error 7.09%, Precision 0.326, Recall 0.856, F1 0.472. This model captures 85.6% of all active cycle power and is best suited for overall energy accounting and billing.
- **Two-Stage Architecture (Classifier + Regressor):** MAE 13.09 W, RMSE 136.80 W, NDE 0.997, Total Energy Error 44.37%, Precision 0.563, Recall 0.710, F1 0.628. By explicitly gating predictions with a binary classifier, it substantially reduces false activations and ghost power draw.

Primary Failure Mode: The main source of false positive predictions is the dishwasher. Both the washing machine and dishwasher employ ~2.2 kW resistive heating elements. When only aggregate active power is measured, a dishwasher heating surge creates an identical step-change in mains demand. Disambiguating these without reactive power (V-I trajectory) data remains a known limitation of single-channel real power NILM.

5. Utility Recommendations & Grid Transferability

- **Demand Response (Time-of-Use):** In House 1, ~30% of wash cycles start during the evening peak window (17:00 to 21:00). Because washing machines are deferrable loads, offering a 30-40% tariff discount for running between 11:00 and 15:00 (solar peak) or overnight could shift 1.5 to 2.0 kW per household off peak feeders.
- **Transferability to Indian Power Grids:** European washers draw ~2 kW primarily because of built-in internal water heating elements. In Indian urban households, washing machines connect directly to cold water inlets and do not heat water internally, drawing only 300-500 W for the motor. Instead, high-power resistive loads in India are driven by separate electric water heaters (geysers: 2-3 kW) and domestic water pumps (0.75-1.5 kW). A model trained on UK data cannot be applied directly to India without recalibrating power thresholds and appliance signatures.
- **Smart Meter Resolution Impact:** REFIT provides 8-second data resampled to 1 minute. Most commercial smart meters roll out at 15-minute or 30-minute intervals. At 15-minute resolution, the 8-minute 2 kW heating spike is averaged down into a minor 250 W elevation, blending into background refrigerator or entertainment loads. High-accuracy cycle start detection requires ≤ 1-minute telemetry; for 15/30-minute meters, models should target aggregate daily energy estimation rather than minute-level event detection.